Smart Quantum AI Engine for Portfolio Optimization

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***Abstract*—Classical portfolio optimization has relied on Harry Markowitz's mean–variance framework since 1952, a model that assumes normally distributed returns and reduces risk to a single second-moment quantity. Empirically, asset returns are asymmetric and heavy-tailed, and portfolio selection under a cardinality constraint is a combinatorial problem that scales poorly for classical solvers once higher-order statistics are introduced. This paper presents a quantum–classical hybrid framework that extends the classical objective to a Mean–Variance–Skewness–Kurtosis (MVSK) formulation, encodes the resulting higher-order objective as a Higher-Order Unconstrained Binary Optimization (HUBO) problem, reduces it to a Quadratic Unconstrained Binary Optimization (QUBO) problem through auxiliary-variable substitution, and solves it with the Quantum Approximate Optimization Algorithm (QAOA) implemented in Qiskit. A Gaussian Hidden Markov Model continuously infers the prevailing market regime – bull, neutral, or bear from rolling volatility and momentum features and dynamically re-weights the four moment terms of the objective, so that the optimizer reflects present market conditions rather than a static historical average. The complete pipeline is benchmarked on five years of daily data across five asset classes (equity, gold, technology equity, cryptocurrency, and long-duration treasuries) against two classical baselines Markowitz mean-variance optimization and simulated annealing on the identical QUBO matrix. Under a moderate risk configuration, the QAOA-based MVSK portfolio achieved a Sharpe ratio of 1.596, closely tracking simulated annealing (1.604) and remaining within reach of the Markowitz baseline (1.734), while retaining the ability to encode skewness and kurtosis preferences that the classical quadratic model cannot express. The results indicate that near-term quantum hardware, through the QAOA–HUBO–QUBO pipeline, can already produce portfolios that are competitive with classical heuristics while supporting a richer and more realistic risk model.**

***Index Terms*—quantum approximate optimization algorithm, portfolio optimization, higher-order moments, skewness, kurtosis, QUBO, HUBO, hidden Markov model, market regime detection, quantum finance**

I. Introduction

Selecting the best combination of assets from a large investment universe, so that expected return is maximized while risk stays within an acceptable bound, is one of the oldest and most computationally demanding problems in quantitative finance. The classical treatment of this problem, due to Markowitz [1], characterizes portfolio risk using only the mean and the variance of asset returns. This two-moment view is elegant and tractable, but it implicitly assumes that returns are normally distributed – an assumption that decades of empirical work have shown to be unrealistic. Real return distributions are skewed and exhibit heavier tails than a Gaussian model predicts, and a portfolio that looks efficient under a mean–variance criterion can be significantly more exposed to extreme downside events than the model suggests.

Two further difficulties compound the distributional problem. First, once an investor is restricted to choosing exactly k assets out of n rather than allocating fractional weight across all of them the selection problem becomes combinatorial, and its complexity grows rapidly with n. Extending the objective with skewness and kurtosis terms turns the mean–variance quadratic program into a higher-order polynomial optimization problem, which is generally NP-hard and quickly outgrows what classical solvers can handle exactly. Second, financial markets are non-stationary they transition between low-volatility bull regimes, neutral conditions, and high-volatility bear regimes, and a static model calibrated on historical data does not adapt to the regime that is actually in force when the portfolio is constructed.

This paper addresses both difficulties within a single deployable system. We formulate portfolio selection as a Mean–Variance–Skewness–Kurtosis (MVSK) objective over a binary asset-selection vector, express it as a Higher-Order Unconstrained Binary Optimization (HUBO) problem, and reduce the HUBO to a Quadratic Unconstrained Binary Optimization (QUBO) problem through auxiliary variable substitution so that it can be solved with the Quantum Approximate Optimization Algorithm (QAOA) [12]. A Gaussian Hidden Markov Model (HMM) infers the current market regime from rolling volatility and momentum and adjusts the relative weighting of the four moments accordingly, so the optimization objective itself changes with market conditions rather than remaining fixed. The complete pipeline data ingestion, statistical preprocessing, regime detection, HUBO construction, QUBO reduction, QAOA solving, and portfolio weight computation is implemented end to end and exposed through an interactive web interface, and it is benchmarked against classical Markowitz optimization and simulated annealing on the same problem instance.

A. Objectives

The specific objectives of this work are to: (i) build a portfolio selection model that incorporates skewness and kurtosis alongside mean and variance rather than relying on the two-moment classical view (ii) integrate a market-regime detector that adapts the optimization weights to prevailing conditions (iii) implement a complete quantum–classical pipeline from raw price data to a recommended allocation, deployable through a browser-based interface and (iv) benchmark the quantum MVSK optimizer against the Markowitz mean–variance optimizer and a simulated-annealing QUBO solver using standard risk-adjusted performance metrics.

B. Scope

The system optimizes an equity-and-multi-asset portfolio using historical daily closing prices obtained through the `yfinance` interface to Yahoo Finance. Portfolio construction is static – performed once for a user-specified horizon and risk tolerance – rather than dynamically rebalanced across multiple periods. Quantum optimization is implemented in Qiskit and can execute on either a local simulator or real IBM Quantum hardware. Live trading execution, transaction-cost modelling beyond the benchmarking layer, and real-time data feeds are outside the present scope.

II. Related Work

Markowitz's mean–variance framework [1] remains the reference point for portfolio theory, but its sensitivity to estimation error and its Gaussian assumption have motivated a long line of refinements. Moon and Yao [2] reformulated the mean-absolute-deviation risk measure as a robust linear program, showing gains over the nominal model that were nonetheless inconsistent across market regimes. DeMiguel et al. [3] showed empirically that none of fourteen optimizing allocation models reliably beat the naive $1/N$ benchmark, tracing the failure to estimation error in expected returns and establishing $1/N$ as a demanding baseline. Rockafellar and Uryasev [4] reformulated risk minimization around Conditional Value-at-Risk, improving tail-risk control at the cost of scenario-generation overhead. Jagannathan and Ma [5] showed that seemingly ad hoc no-short-selling constraints act as an implicit shrinkage estimator on the covariance matrix, stabilizing portfolios without addressing the underlying model misspecification. None of these classical refinements incorporates skewness and kurtosis directly into a combinatorial selection framework, which is the gap the present work targets.

On the quantum side, Rebentrost and Lloyd [6] proposed one of the first quantum algorithms for portfolio optimization, using a quantum linear-systems solver to obtain the efficient frontier in time logarithmic in the size of the historical return dataset, at the cost of relying on quantum random access memory whose physical realization remains an open engineering problem. Marzec [9] used a D-Wave adiabatic annealer to solve a graph-theoretic asset-selection formulation, assigning equal weight to all selected assets rather than optimizing continuous weights. Grant, Humble, and Stump [10] used portfolio optimization as a benchmark to study D-Wave 2000Q annealing controls across a thousand random QUBO instances, finding that reverse annealing seeded from a forward-annealing solution improved success probability by roughly an order of magnitude – a result that directly motivated our practice of running the QAOA solver multiple times per instance and keeping the best solution. Mugel et al. [7] tackled dynamic, transaction-cost-aware portfolio optimization at a much larger scale (up to 1272 qubits) using D-Wave hybrid annealing, IBM-Q variational solvers, and quantum-inspired tensor networks, but did not extend the objective beyond mean and variance. Most recently, Innan et al. [11] systematically benchmarked QAOA and Sampling-VQE with several ansatz families on real 2025 market data and introduced an expert-evaluation layer, finding that deeper circuits do not automatically translate into more financially sound portfolios – a finding that reinforced our decision to select the QAOA run with the best realized Sharpe ratio rather than the run with the lowest raw QUBO energy.

The work closest to ours is Uotila, Ripatti, and Zhao [8], who developed the first quantum formulation of portfolio optimization that includes skewness and kurtosis, producing a HUBO problem solved with QAOA and integer-variable encoding, and showing that the HUBO formulation frequently yields more diversified portfolios than mean-variance QUBO baselines, at the cost of degraded QAOA performance beyond roughly ten qubits. Our system implements the same MVSK-HUBO-QUBO pipeline and extends it with a Hidden Markov Model regime detector that dynamically re-weights the four moment terms, which to our knowledge has not been combined with a higher-moment quantum objective in prior work.

III. Problem Statement

Investors must choose a subset of assets from a universe of candidates, each with different risk and return characteristics. The classical Markowitz model simplifies this choice by assuming Gaussian returns and using variance as the sole measure of risk; in practice, returns are skewed and prone to extreme moves that variance does not capture, so mean-variance-optimal portfolios can perform poorly precisely when protection is most needed. In addition, markets shift between bull, neutral, and bear regimes, and a static optimization objective that ignores these shifts will systematically misallocate capital relative to current conditions. This paper addresses both limitations by folding all four statistical moments into the optimization objective and by adapting that objective to the market regime detected by a Hidden Markov Model. Because the resulting higher-order combinatorial problem is intractable for classical exact solvers as the asset universe grows, the Quantum Approximate Optimization Algorithm is used to obtain near-optimal allocations.

IV. Proposed System

The proposed system is a quantum–classical hybrid pipeline with three central design choices. First, the optimization objective is extended from mean–variance to a full Mean–Variance–Skewness–Kurtosis objective, encoded as a HUBO over a binary asset-selection vector. Second, a Gaussian HMM fitted on rolling volatility and momentum features infers the prevailing market regime and rescales the four moment-weighting parameters accordingly, so the optimizer solves a formulation that reflects present conditions rather than a fixed historical average. Third, the resulting QUBO (after HUBO reduction) is solved with QAOA in Qiskit, with support for both local simulation and execution on IBM Quantum hardware.

The system accepts three inputs from the investor – an investment budget, a risk tolerance on a 1–10 scale, and an investment horizon in days – and returns a recommended allocation together with a benchmark comparison table against Markowitz mean–variance optimization and simulated annealing on the identical QUBO matrix, reporting mean return, volatility, Sharpe ratio, and maximum drawdown for all three methods. The entire pipeline is exposed through a Streamlit web application, so that the underlying quantum optimization is usable without requiring the end user to understand quantum computing.

V. Mathematical Formulation

A. Binary Asset Selection

Let there be n candidate assets indexed by $i \in \{1, \dots, n\}$, each with a historical log-return series. The selection decision is encoded as a binary vector $x \in \{0, 1\}^n$, where $x_i = 1$ indicates that asset i is included in the portfolio. The number of assets to be selected, k, is derived from the investor's risk tolerance $r \in [1, 10]$ as

$$k = \min\Big(\Big\lfloor 2 + \frac{r}{4} \Big\rfloor,\ n\Big), \tag{1}$$

so that higher risk tolerance yields a larger, more diversified selection, and the cardinality constraint is $\sum_{i=1}^{n} x_i = k$.

B. MVSK Objective

Let $\mu \in \mathbb{R}^n$ denote the mean log-return vector, $\Sigma^- \in \mathbb{R}^{n\times n}$ the downside covariance matrix (computed after zeroing all positive returns, so that only loss-generating co-movement contributes to the risk term), S the co-skewness tensor, and K the co-kurtosis tensor. The portfolio objective, written as a minimization, is

$$\min_x\ -\lambda_{\text{ret}}\, \mu^\top x \ +\ \lambda_{\text{var}}\, x^\top \Sigma^- x \ -\ \lambda_{\text{skew}}\, S(x) \ +\ \lambda_{\text{kurt}}\, K(x), \tag{2}$$

where $S(x)$ and $K(x)$ denote the skewness and kurtosis contributions of the selected assets. The four coefficients $\lambda_{\text{ret}}, \lambda_{\text{var}}, \lambda_{\text{skew}}, \lambda_{\text{kurt}}$ are not fixed constants: they are first set from the investor's risk-aversion coefficient

$$\gamma = \frac{11 - r}{10}, \tag{3}$$

giving base values $\lambda_{\text{ret}} = 10$, $\lambda_{\text{var}} = 200\,\gamma$, $\lambda_{\text{skew}} = 50(1-\gamma)$, $\lambda_{\text{kurt}} = 150\,\gamma$, and are then rescaled by the detected market regime: in a bear regime λ_{var} is doubled and λ_{kurt} increased by 50%, while in a bull regime λ_{ret} and λ_{skew} are scaled up by 25% and $\lambda_{\text{var}}, \lambda_{\text{kurt}}$ are reduced by 10%.

C. Moment Estimation

The mean vector is the column-wise sample mean of the log-return matrix $R \in \mathbb{R}^{T\times n}$, $\mu = \frac{1}{T}\sum_t R_{t,\cdot}$. The downside covariance is computed from a return matrix R^- in which all positive entries are set to zero, $\Sigma^- = \mathrm{Cov}(R^-)$, so it isolates co-movement during loss-generating periods. Working from the demeaned return series $\tilde{R} = R - \mu$, the co-skewness and co-kurtosis tensors are built as fully symmetric third- and fourth-order tensors, respectively,

$$S_{ijk} = \frac{1}{T}\sum_{t=1}^{T} \tilde{R}_{t,i}\, \tilde{R}_{t,j}\, \tilde{R}_{t,k}, \qquad K_{ijkl} = \frac{1}{T}\sum_{t=1}^{T} \tilde{R}_{t,i}\, \tilde{R}_{t,j}\, \tilde{R}_{t,k}\, \tilde{R}_{t,l}. \tag{4}$$

Only the fully diagonal elements S_{iii} and K_{iiii} are retained in the HUBO objective, which keeps the resulting cubic and quartic contributions linear in x and makes the problem computationally tractable at the qubit counts reachable on near-term hardware.

D. HUBO Construction

Collecting the linear return term, the quadratic downside-variance term, the (diagonal) skewness and kurtosis corrections, and a quadratic penalty enforcing the cardinality constraint, the HUBO objective is

$$\begin{aligned} H(x) = &\sum_i \Big(-\mu_i\lambda_{\text{ret}} - \lambda_{\text{skew}} S_{iii} + \lambda_{\text{kurt}} K_{iiii} \Big) x_i \\ &+ \lambda_{\text{var}} \sum_{i,j} \Sigma^-_{ij}\, x_i x_j \ +\ P\Big(\sum_i x_i - k\Big)^2, \end{aligned} \tag{5}$$

where P is a penalty coefficient set proportional to the largest magnitude among μ and Σ^- (via $P = 10 \cdot \max(\|\mu\|_\infty, \|\Sigma^-\|_\infty)$ in our implementation), which prevents the cardinality penalty from either dominating or being dominated by the objective terms. Expanding the quadratic penalty contributes a linear term $P(1 - 2k)$ per asset and a pairwise cross term $2P$ for every pair (i, j), $i \neq j$; the constant Pk^2 term is dropped, as it does not affect the arg-min.

E. HUBO to QUBO Reduction

QAOA requires a purely quadratic cost function. Any term of degree higher than two – which can arise from the cross terms introduced when reducing multi-asset products – is eliminated by auxiliary-variable substitution: for a product $x_i x_j$ appearing inside a higher-degree term, a new binary variable x_{aux} is introduced with a penalty

$$\rho\big(3x_{aux} - 2x_i x_{aux} - 2x_j x_{aux} + x_i x_j\big), \tag{6}$$

which is minimized exactly when $x_{aux} = x_i x_j$, so that the substitution is enforced at the optimum. After every such reduction the resulting expression is purely quadratic and is assembled into a dense symmetric matrix $Q \in \mathbb{R}^{m\times m}$ (with $m \geq n$ once auxiliary variables are counted), so that the full problem is $x^\top Q x$.

F. Ising Encoding and QAOA

To solve $\min_x x^\top Q x$ with QAOA, binary variables are mapped to spin variables via $x_i = (1 - z_i)/2$, $z_i \in \{-1, +1\}$, transforming the QUBO into an Ising Hamiltonian

$$H_C = \sum_i h_i Z_i + \sum_{i<j} J_{ij} Z_i Z_j, \tag{7}$$

with $h_i = Q_{ii}/2$ and $J_{ij} = Q_{ij}/4$ derived directly from the entries of Q. Each qubit corresponds to one decision variable x_i. The QAOA ansatz alternates p layers of the cost unitary $e^{-i\gamma_l H_C}$ and the transverse-field mixer $e^{-i\beta_l H_M}$, $H_M = \sum_i X_i$, and the $2p$ variational parameters (γ_l, β_l) are optimized classically to minimize $\langle\psi(\gamma,\beta)|H_C|\psi(\gamma,\beta)\rangle$. The resulting state is measured in the computational basis; the sampled bitstring of lowest QUBO energy is decoded back into the asset-selection vector x.

G. Weight Computation

Given the selected assets, continuous portfolio weights are obtained by a restricted mean–variance sub-optimization: with $\mu_{\mathcal{S}}$ and $\Sigma_{\mathcal{S}}$ the mean vector and covariance matrix restricted to the selected index set $\mathcal{S}$, weights are computed as $w \propto \max\left(\Sigma_{\mathcal{S}}^{-1}\mu_{\mathcal{S}},\, 0\right)$, clipped at zero to exclude short positions and normalized to sum to one; each weight is then scaled by the investor's budget to obtain a monetary allocation.

VI. System Architecture and Implementation

A. Pipeline Overview

The backend is implemented in Python and executes sequentially through six stages: data ingestion and preprocessing, feature computation, market-regime detection, HUBO construction, HUBO-to-QUBO reduction, and quantum/classical solving, followed by weight computation and benchmarking. Each stage is an independently testable module, and the same pipeline is invoked from both a command-line entry point and a Streamlit web front end.

B. Data and Preprocessing

The processed log-return matrix is loaded and filtered to the user-specified horizon with a trailing window. The mean vector and full covariance matrix are computed from the filtered window and normalized by their maximum absolute values before being used in the HUBO, a step found empirically necessary: unnormalized values produced ill-conditioned QUBO matrices that drove the QAOA solver toward degenerate solutions.

C. Market Regime Detection

Regime inference uses a three-state Gaussian HMM with diagonal covariance, trained for 200 iterations on a pair of standardized, cross-asset-averaged rolling statistics: a 30-day rolling volatility and a 30-day rolling momentum term. The most recent hidden state produced by the model is mapped to regime 0 (bull), 1 (neutral), or 2 (bear), which in turn rescales the four λ coefficients of the MVSK objective as described in Section V-B; the optimizer therefore tracks prevailing conditions automatically, without the investor having to specify a regime by hand.

D. QAOA Solver Configuration

The QAOA circuit is built with Qiskit's `QAOAAnsatz` at a circuit depth of 3–4 layers. Circuit parameters are optimized with a classical variational optimizer (SPSA in the production configuration, with SLSQP retained as an earlier baseline), starting from a fixed initial parameter vector for reproducibility, and the QUBO matrix is normalized by its largest-magnitude entry before conversion to an Ising Hamiltonian to keep the optimization landscape well conditioned. On the local `AerSimulator` backend the statevector is available exactly, so the final measurement distribution is obtained by inspecting the top-probability basis states directly rather than by finite-shot sampling; the solver is executed five times per query and the run with the highest realized Sharpe ratio is retained, following the observation in [11] that circuit convergence does not guarantee financial soundness. When routed to real IBM Quantum hardware, the circuit is measured with a fixed shot budget and a single run is performed, with the most frequent bitstring taken as the solution.

E. Classical Baselines

Simulated annealing is run on the identical QUBO matrix using a geometric cooling schedule, so its result is directly comparable to the quantum solution on exactly the same problem instance. The Markowitz baseline solves the classical Sharpe-ratio-maximizing quadratic program with SciPy's SLSQP optimizer under non-negativity and sum-to-one constraints, independent of the QUBO formulation.

F. Dataset

The evaluation dataset consists of daily adjusted closing prices for five assets spanning distinct asset classes: SPY (broad US equity), GOLD (SPDR Gold Shares, a commodity/safe-haven proxy), AAPL (large-cap technology equity), BTC (Bitcoin, representing digital assets), and TLT (long-duration US Treasury bonds), downloaded via the `yfinance` interface to Yahoo Finance and covering the period from September 2014 to February 2026, yielding 2,236 daily log-return observations per asset. This combination was chosen specifically because it spans equity, commodity, cryptocurrency, and fixed-income risk profiles, so that correlation structure and higher-moment behaviour differ meaningfully across assets – a necessary condition for the MVSK formulation to add value over a mean–variance model. Gold exhibited the highest observed volatility and the most extreme single-day loss in the sample, Bitcoin exhibited the highest single-day gain and mean return, and Treasuries showed the lowest mean return and volatility, consistent with their role as a defensive asset; this spread confirms that the return distributions in the dataset are non-Gaussian and motivates the inclusion of skewness and kurtosis terms.

VII. Results and Discussion

We swept the budget, risk-tolerance, and horizon inputs across a range of settings and, in each case, compared the QAOA-based MVSK optimizer against the Markowitz baseline and against simulated annealing run on that instance's exact QUBO matrix. Table I reports one representative comparison, taken at a moderate risk tolerance of 5 with a 200-day horizon.

Under this configuration, the quantum MVSK portfolio achieved a Sharpe ratio of 1.596, within 0.5% of simulated

TABLE I
BENCHMARK COMPARISON UNDER A MODERATE-RISK CONFIGURATION (RISK = 5, HORIZON = 200 DAYS)

Method	Return	Vol.	Sharpe	Max DD
Quantum MVSK (QAOA)	0.224	0.140	1.596	−0.062
Simulated Annealing	0.236	0.147	1.604	−0.069
Markowitz	0.201	0.116	1.734	−0.047

annealing on the same QUBO instance (1.604) and roughly 8% below the unconstrained continuous-weight Markowitz solution (1.734). This gap is expected: the Markowitz baseline optimizes continuous weights directly over the full asset universe, whereas the QAOA and simulated-annealing solutions are additionally constrained to a discrete cardinality-k selection before weights are computed, which is a strictly harder combinatorial problem. Under higher-risk configurations the quantum model produced returns of 0.317 (Sharpe 1.48) at risk level 10 and up to 0.333 at risk level 8, indicating that the optimizer continues to identify favorable asset combinations as the cardinality and return-weighting parameters change.

Performance was not uniform across all configurations: at risk level 5 with a 250-day horizon, the quantum solution's Sharpe ratio dropped to 0.096, an outcome attributable to the sensitivity of the variational optimization landscape to input conditions – specifically, changes to the covariance normalization and cardinality target under the longer horizon shifted the QUBO energy landscape enough that the retained QAOA run converged to a comparatively weak local optimum despite the best-of-five-runs selection strategy. This variability is consistent with the observation in [11] that QAOA convergence and portfolio quality are not always aligned, and it motivates the best-of-several-runs strategy already used in our solver as a partial mitigation.

Taken together, the results show that the quantum MVSK approach does not consistently outperform classical methods in raw risk-adjusted return, but it produces solutions that are competitive with simulated annealing on the identical combinatorial problem and remain within a modest margin of the unconstrained classical benchmark, while additionally supporting skewness and kurtosis preferences and regime-adaptive weighting that the classical Markowitz baseline cannot express by construction. This is broadly consistent with the findings of Uotila et al. [8], who likewise report that HUBO-based quantum portfolios are competitive with, rather than uniformly superior to, classical baselines at problem sizes reachable on current simulators and near-term hardware.

VIII. CONCLUSION AND FUTURE WORK

This paper presented a complete quantum–classical hybrid pipeline for portfolio optimization that extends the classical Markowitz objective to incorporate skewness and kurtosis, encodes the resulting higher-order objective as a HUBO that is reduced to a QUBO and solved with QAOA, and adapts the optimization weights to the market regime detected by a Gaussian Hidden Markov Model. Benchmarked against Markowitz mean–variance optimization and simulated annealing on five years of multi-asset daily data, the quantum MVSK optimizer produced portfolios that were competitive with simulated annealing on the identical combinatorial problem and within a modest margin of the classical continuous-weight baseline, while exposing a risk model that the classical baseline cannot represent. The results also surfaced a sensitivity of QAOA convergence to input conditions and problem scaling, consistent with prior benchmarking work in the field.

Future work includes extending the co-skewness and co-kurtosis tensors beyond their diagonal elements to capture cross-asset higher-moment interactions, evaluating the pipeline at larger asset universes and qubit counts on real IBM Quantum hardware rather than simulation alone, incorporating transaction costs and dynamic multi-period rebalancing following the direction of [7], and replacing the current best-of-five-runs heuristic with a warm-start or counterdiabatic initialization strategy to reduce the sensitivity to the variational landscape observed in some configurations.

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REFERENCES

[1] H. Markowitz, "Portfolio selection," *The Journal of Finance*, vol. 7, no. 1, pp. 77–91, 1952.
[2] Y. Moon and T. Yao, "A robust mean absolute deviation model for portfolio optimization," *Computers & Industrial Engineering*, vol. 40, no. 5, pp. 1245–1254, 2011.
[3] V. DeMiguel, L. Garlappi, and R. Uppal, "Optimal versus naive diversification: How inefficient is the 1/N portfolio strategy?" *The Review of Financial Studies*, vol. 22, no. 5, pp. 1915–1953, 2009.
[4] R. T. Rockafellar and S. Uryasev, "Optimization of conditional value-at-risk," *Journal of Risk*, vol. 2, no. 3, pp. 21–41, 2000.
[5] R. Jagannathan and T. Ma, "Risk reduction in large portfolios: Why imposing the wrong constraints helps," *The Journal of Finance*, vol. 58, no. 4, pp. 1651–1683, 2003.
[6] P. Rebentrost and S. Lloyd, "Quantum computational finance: quantum algorithm for portfolio optimization," *arXiv preprint arXiv:1811.03975*, 2018.
[7] S. Mugel, C. Kuchkovsky, E. Sánchez, S. Fernández-Lorenzo, J. Luis-Hita, E. Lizaso, and R. Orús, "Dynamic portfolio optimization with real datasets using quantum processors and quantum-inspired tensor networks," *Physical Review Research*, vol. 4, p. 013006, 2022.
[8] V. Uotila, J. Ripatti, and B. Zhao, "Higher-order portfolio optimization with quantum approximate optimization algorithm," *arXiv preprint arXiv:2509.01496*, 2025.
[9] M. Marzec, "Portfolio optimization: Applications in quantum computing," in *Handbook of High-Frequency Trading and Modeling in Finance*, Wiley-Blackwell, 2016, pp. 73–106.
[10] E. Grant, T. S. Humble, and B. Stump, "Benchmarking quantum annealing controls with portfolio optimization," *Physical Review Applied*, vol. 15, p. 014012, 2021.
[11] N. Innan, A. Saleem, A. Marchisio, and M. Shafique, "Quantum portfolio optimization with expert analysis evaluation," *arXiv preprint arXiv:2507.20532*, 2025.
[12] E. Farhi, J. Goldstone, and S. Gutmann, "A quantum approximate optimization algorithm," *arXiv preprint arXiv:1411.4028*, 2014.